

# Python and Table Readiness

Project: Responsible Plant Classifier

## 1. Python and Table Readiness

### Teach the learning cycle

Primary target: Specify a table schema

#### Session · Sequence and minutes · Resources

**Session:** Session 1 · Understand · 45 min

**Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

**Resources:** Sections 1–6; record the brief recall in the workbook.

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**Session:** Session 2 · Practise · 45 min

**Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

**Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

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**Session:** Session 3 · Investigate and transfer · 45 min

**Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

**Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

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This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building. Repeated copies of a case share one recorded attempt; they are not new evidence.

Case exposure check: this source example appears in ai-2/7, ai-4/4. Reuse a familiar case for recall or

modelling; verify that the case used to assess transfer has not already received feedback.

## Retrieval answer guidance

### Classification or Regression?

Without opening the old lesson, explain one key idea from Classification or Regression?. Give an example and a mistake that the idea helps you avoid.

Look for: Identify a task from the meaning of its target

Reference example, not the only acceptable wording: Route type is classification because 1 and 2 encode categories. Whole-minute duration is regression because numerical differences measure time. Exceeding 10 minutes is binary classification because the target is yes or no.

### The ML Lifecycle

Without opening the old lesson, explain one key idea from The ML Lifecycle. Give an example and a mistake that the idea helps you avoid.

Look for: Distinguish fitting a rule from applying a fixed rule

Reference example, not the only acceptable wording: The input is width  $x$  and the label is the observed category 0 or 1. Candidates are 2, 4 and 6. Selection uses mistakes on the four training cards only; held out labels must not guide it.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

### What the learner must demonstrate

Trace mean for [4,8,12] using  $\text{sum}(\text{values})/\text{len}(\text{values})$ . What should the program do for []? Explain why returning 0 silently is misleading.

### Concept to explain

Python variables, types, expressions, rows, and columns give a reproducible representation of data and transformations.

#### Term · Meaning

**Term:** variable

**Meaning:** name bound to a value

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**Term:** data type

**Meaning:** kind of value and valid operations

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**Term:** dataframe

**Meaning:** table represented in code

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**Term:** expression

**Meaning:** code that computes a value

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## Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

## Preparation for the connected explanation

Teach Week 13 as a two-session investigation using fictional data. A fictional classroom logs the duration of each activity. Before comparing durations, define a schema: activity is nonempty text, minutes is an integer from 0 through 60, and each row describes one activity session. These decisions make data interpretable. Assess the three stated outcomes through the independent task, not by copying the worked result.

## Readiness answer

Look up the value associated with the key "used" in that record.

Python 3 editor with a fresh local practice file

Paper trace table and the student workbook

Fictional input records from this lesson; no real student identifiers

## Apply the connected idea: File input and output

Teach Week 25 as a two-session investigation using fictional data. Use a teacher-provided local sandbox folder, never a real personal file. Create durations.txt with two lines: 12 and 8. The goal is to read the text, convert it to integers and report the total. Keep the original file unchanged for this activity. Assess the three stated outcomes through the independent task, not by copying the worked result.

## Readiness answer

It is text; conversion and validation are needed.

Python 3 editor with a fresh local practice file

Paper trace table and the student workbook

Fictional input records from this lesson; no real student identifiers

### Apply the connected idea: Exceptions and recovery

Teach Week 27 as a two-session investigation using fictional data. A small parser receives duration text. Its contract permits a string representing a nonnegative integer up to 60. Invalid numeric text should return a clear status rather than crash or pretend that the duration was zero. Test both the successful and failing paths. Assess the three stated outcomes through the independent task, not by copying the worked result.

### Readiness answer

ValueError.

Python 3 editor with a fresh local practice file

Paper trace table and the student workbook

Fictional input records from this lesson; no real student identifiers

### End goal for the pictorial case

By the end, I can explain python and table readiness using a traced example and a stated limitation.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

## 2. Explain the mechanism

### Explain the mechanism

Trace the information, fitted behavior, and evaluation proof separately.

QUESTION Trace a short notebook state and repair two type or missing-field errors.

MODEL IDEA Python variables, types, expressions, rows, and columns give a reproducible representation of data and transformations.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD `widths = [4.2, 5.0, 6.1]` stores three decimal measurements. `mean_width = sum(widths) / len(widths)` evaluates to 5.1.

Draw a fresh data → model → prediction → evaluation trace:

Predict first. Then check each step. A correct answer without visible evidence cannot be audited.

STEP 1 `widths = [4.2, 5.0, 6.1]` stores three decimal measurements.

STEP 2 `mean_width = sum(widths) / len(widths)` evaluates to 5.1.

STEP 3 Adding 'unknown' to the numeric list causes a type problem.

STEP 4 Clean or explicitly encode values, rerun top to bottom, and record the result.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the evidence independently:

## Concrete teaching case

Python `lengths=[4,6,8]; total=sum(lengths); mean=total/len(lengths)`.

Trace and handle an empty list.

## Exact explanation to model

`Total=18, count=3, mean=6.0`. Guard the empty case against division by zero; no mean exists without observations.

## Explain the added worked example

A fictional classroom logs the duration of each activity. Before comparing durations, define a schema: activity is nonempty text, minutes is an integer from 0 through 60, and each row describes one activity session. These decisions make data interpretable.

The worked script prints True twice. `type(value) is int` checks the exact type here and avoids accepting a Boolean as a count. The chained comparison checks both boundaries. The example assumes both keys exist; next week handles missing fields. A zero duration is allowed by this written schema.

## Key terms

### Term · Meaning

**Term:** Schema

**Meaning:** Rules for the fields and values in records

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**Term:** Type

**Meaning:** A category such as integer or text

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**Term:** Unit

**Meaning:** The quantity scale attached to a number

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**Term:** Constraint

**Meaning:** A rule a valid record must satisfy

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### Apply the connected idea: File input and output

Use a teacher-provided local sandbox folder, never a real personal file. Create durations.txt with two lines: 12 and 8. The goal is to read the text, convert it to integers and report the total. Keep the original file unchanged for this activity.

The mode `r` means read. `with` closes the file when the block finishes, including when an error occurs. `splitlines` separates lines without retaining newline characters. This example assumes each line is a valid integer text; an empty line or twelve raises `ValueError` during `int` conversion.

Values `[0, 3, 7]`, total 10. Exit: It is a parse failure, not a measured zero. Flag the line and report the problem; silently substituting changes the evidence.

### Apply the connected idea: Exceptions and recovery

A small parser receives duration text. Its contract permits a string representing a nonnegative integer up to 60. Invalid numeric text should return a clear status rather than crash or pretend that the duration was zero. Test both the successful and failing paths.

`try` covers the operation expected to fail; except `ValueError` handles that known conversion failure. Returning early ends the failing path. This contract expects a string: it does not claim to handle every possible object. Spaces around an integer are accepted by `int`, and that behavior should be documented.

out of range/`None`, ok/5, invalid text/`None`. Exit: Accepted count 8, rejected count 2 and reasons; summaries apply to accepted data. Preserve raw input for authorized correction.

### Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a prediction, then explain the deciding step. These teacher notes contain solutions; do not reveal the independent case answer during modelling.

**Would minutes of 61, "12" and 12 each pass? Explain the type distinction.**

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## Thinking move · Teacher prompt

**Thinking move:** Plan

**Teacher prompt:** What is given? What is the question? Which procedure fits these facts?

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**Thinking move:** Do

**Teacher prompt:** Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.

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**Thinking move:** Check

**Teacher prompt:** Does the output answer the question? Which assumption or limitation prevents a larger claim?

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Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

## A pictorial worked case: Python and Table Readiness

Inspect a small table before asking Python to compute an average.

### The facts we will use

Plant heights in cm: P=2, Q=6, R=None, S=0. None means unmeasured; zero is a recorded valid height.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

## Python and Table Readiness

### Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



#### 1. Check records

Four rows contain three measured values and one missing value.



#### 2. Calculate mean

Measured sum is  $2+6+0=8$ ; mean is  $8/3$  cm, approximately 2.67 cm.



#### 3. Report coverage

Report 3/4 measured, retaining the missing record separately.

### CHANGE ONE THING

S changes from 0 to None.

Explain the changed result using the matching lesson.

### Read the picture one step at a time

#### Step · What happens and why

**Step:** 1. Check records

**What happens and why:** Four rows contain three measured values and one missing value.

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**Step:** 2. Calculate mean

**What happens and why:** Measured sum is  $2+6+0=8$ ; mean is  $8/3$  cm, approximately 2.67 cm.

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### Step: 3. Report coverage

**What happens and why:** Report 3/4 measured, retaining the missing record separately.

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Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

### Why this result follows

Read the picture in order. Report 3/4 measured, retaining the missing record separately. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

### What this example does not establish

The measured-row mean may not represent the missing plants; investigate why those readings are absent.

### Change one thing in the pictured case

Change: S changes from 0 to None.

Prediction: Mean becomes  $(2+6)/2=4$  cm with 2/4 measured.

Reason: Removing a measured zero changes both the denominator and the coverage.

### Ask while showing the picture

Cover the result. Ask learners to name the inputs and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its evidence. Then change the supplied condition and compare. The picture is a representation; it does not document execution of a real AI model.

## 3. Demonstrate and guide practice

### Demonstrate and guide practice

#### Board sequence

Write the given facts: Python `lengths=[4,6,8]`; `total=sum(lengths)`; `mean=total/len(lengths)`.

Write the question to explain: Trace and handle an empty list.

Reveal after predictions: `Total=18`, `count=3`, `mean=6.0`. Guard the empty case against division by zero; no mean exists without observations.

Keep for an independent attempt: Trace Python: `lengths = [4, 7, 9]`; `selected = lengths[1]`; `total = selected + 3`. Give selected and total.

### Classroom dialogue

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks evidence; exchange roles. Remove the worked answer before independent assessment.

## Model these guided questions

61 fails the range; "12" fails the integer type; 12 passes. Exit: No. The value has a permitted structure but still needs trustworthy collection evidence.

## Case-specific teaching sequence

### Stage · Teacher action

**Stage:** Session 1 start to 7 minutes

**Teacher action:** Retrieve the prerequisite and explain the purpose in the opening section.

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**Stage:** 7 to 20 minutes

**Teacher action:** Read the worked example and diagram; ask learners to trace each step aloud.

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**Stage:** 20 to 33 minutes

**Teacher action:** Complete W1 and W2, then compare predictions with observed results.

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**Stage:** 33 to 45 minutes

**Teacher action:** Collect the guided prediction before running; discuss a mismatch and its cause.

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**Stage:** Session 2 start to 8 minutes

**Teacher action:** Retrieve the main rule without showing the previous answer.

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**Stage:** 8 to 20 minutes

**Teacher action:** Use W3 to diagnose and repair the stated failure; rerun an earlier case.

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**Stage:** 20 to 35 minutes

**Teacher action:** Collect W4 independently, including code or plan, predictions and explanations, before hints.

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**Stage:** 35 to 45 minutes

**Teacher action:** Discuss W5 and the exit; use the fresh retry when support was needed.

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## Teach the connected idea

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## Reduce help deliberately

Completion task: Classify minutes values 0, 60, -1 and 61 under this schema.

Teacher check: 0 and 60 pass; -1 and 61 fail.

Start by identifying the given inputs together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood. For partners, alternate explainer and checker; collect a response from every learner before discussion.

## Individual reasoning check

Use the worked case for Rows columns and schemas. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: A row can satisfy every schema rule and still be a wrong observation. Schema validation checks our stated structure and constraints, not whether someone measured honestly or selected a representative sample.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class. If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

## Connected teaching cycle: File input and output

Target: Read a small local data file

Reserve further sessions for this topic; the 135-minute primary plan does not absorb all added content.

Completion practice: A sandbox file has lines 5, 0 and 2. What total does the worked reader print?

Teacher check: The parsed values sum to  $5 + 0 + 2 = 7$ .

Individual check: For File input and output, use its worked case to evaluate this proposed explanation: “A successful run proves the conclusion”. Explain the deciding step.

Expected correction: File reading does not establish that the values are valid durations. After conversion, apply the schema and record any rejected lines. Do not upload personal records to an external tool for this exercise.

Apply the same support, independent transfer and delayed-review decisions to the named connected-topic workbook tasks.

## Connected teaching cycle: Exceptions and recovery

Target: Handle expected exceptions specifically

Reserve further sessions for this topic; the 135-minute primary plan does not absorb all added content.

Completion practice: Compare `parse_minutes("abc")` and `parse_minutes("70")`.

Teacher check: `abc` gives invalid text/None; `70` gives out of range/None. Both fail but for different reasons.

Individual check: For Exceptions and recovery, use its worked case to evaluate this proposed explanation: "A successful run proves the conclusion". Explain the deciding step.

Expected correction: Recovery must fit the task. A caller may ask for corrected input or exclude the row with a recorded reason. It must not use None as a number or claim that rejected records were successfully processed.

Apply the same support, independent transfer and delayed-review decisions to the named connected-topic workbook tasks.

### Pictorial practice question

Why should the missing value not silently become zero? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

### Reasoned teacher answer

Zero is a real recorded measurement here. Replacing missing with zero would invent evidence and lower the mean without justification. Changed-case check: Mean becomes  $(2+6)/2=4$  cm with  $2/4$  measured. Removing a measured zero changes both the denominator and the coverage.

Changed-case reasoning: Mean becomes  $(2+6)/2=4$  cm with  $2/4$  measured. Removing a measured zero changes both the denominator and the coverage.

Collect each learner's explanation before discussion. Use the first missing connection between input, deciding step and result to choose support. This question is worked-case practice, not fresh mastery evidence.

## 4. Run the weekly evidence lab

### Run the weekly evidence lab

Keep the test conditions fixed, record every result, and change only one planned factor.

1. Write the hypothesis and expected evidence before running.
2. Notebook Detective: predict five variable values, trace two table filters, repair two errors, and rerun from a clean state.
3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.
5. Interpret what changed, what did not, and what remains unsafe or unknown.

## Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing. Keep the expected result separate from an observed mistake.

### Record · What to capture

**Record:** Input or evidence

**What to capture:** Exact data, instruction or claim

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**Record:** Expected result

**What to capture:** Answer or safe fallback declared before the run

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**Record:** Observed result

**What to capture:** Actual result, including errors

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**Record:** Explanation

**What to capture:** Rule, calculation or evidence explaining the outcome

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**Record:** Next test

**What to capture:** One justified change and a new prediction

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## Diagnose and reteach

MISCONCEPTION TO REPAIR If notebook cells eventually run, execution order and data types do not matter.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation. Finish with a different uncoached case before recording mastery.

## Address these additional misconceptions

A successful run proves the conclusion: A row can satisfy every schema rule and still be a wrong observation. Schema validation checks our stated structure and constraints, not whether someone measured honestly or selected a representative sample.

Matching the example is enough: Require the new W4 case, explicit predictions and an explanation. Fresh retry: define a tray label and integer count 0 through 20. Require tests at 0, 20, -1 and 21 and a text count.

A convenient shortcut preserves meaning: If one row records 120 seconds in the minutes field, a simple numeric total becomes misleading. Correct the source or explicitly convert and document it; renaming a column alone does not convert values.

### Check these misconceptions

A successful run proves the conclusion: File reading does not establish that the values are valid durations. After conversion, apply the schema and record any rejected lines. Do not upload personal records to an external tool for this exercise.

Matching the example is enough: Require the new W4 case, explicit predictions and an explanation. Fresh retry: sandbox values 2, 7, 1, 0 give count 4 and total 10. An empty file yields count 0 and sum 0, which must be labelled no observations.

A convenient shortcut preserves meaning: If Python cannot find durations.txt, it raises FileNotFoundError. Check the current working folder and exact filename. Do not switch to write mode to solve a missing input: opening an existing file with w would replace its contents.

### Check these misconceptions

A successful run proves the conclusion: Recovery must fit the task. A caller may ask for corrected input or exclude the row with a recorded reason. It must not use None as a number or claim that rejected records were successfully processed.

Matching the example is enough: Require the new W4 case, explicit predictions and an explanation. Fresh retry: integer score strings 0 through 5. "5" ok/5, "6" out of range/None, "" invalid text/None, "0" ok/0.

A convenient shortcut preserves meaning: A bare except that returns 0 hides unrelated programming errors and corrupts missing values. Catch the expected exception, preserve an error status and keep the unknown value as None. If file reading is added, handle FileNotFoundError separately with an actionable message.

### Match the support to the first error

#### What the response shows · Next teaching action

**What the response shows:** Missing or misread facts

**Next teaching action:** Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.



**What the response shows:** Wrong procedure

**Next teaching action:** Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

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**What the response shows:** Arithmetic or implementation error

**Next teaching action:** Trace intermediate values or lines; preserve the first mismatch and retry that step.

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**What the response shows:** Correct result without a reason

**Next teaching action:** Ask for a trace and a changed case before marking independent understanding.

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**What the response shows:** An unsupported wider claim

**Next teaching action:** Ask which supplied evidence supports the claim and which further observation would be needed.

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Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal. Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

## 5. Assess and explain the answers

### Assess and explain the answers

#### Worked case answer

Total=18, count=3, mean=6.0. Guard the empty case against division by zero; no mean exists without observations.

#### Independent question

Trace Python: lengths = [4, 7, 9]; selected = lengths[1]; total = selected + 3. Give selected and total.

#### Expected independent answer

Python indexes this list from zero: lengths[1] is 7, so selected is 7 and total is 10.

#### What counts as evidence

Criterion · 0 points · 1 point · 2 points

### Criterion: Result

**0 points:** Incorrect or absent

**1 point:** Partly correct

**2 points:** Correct with necessary conditions

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### Criterion: Reasoning

**0 points:** No supporting trace

**1 point:** Some steps

**2 points:** Clear mechanism, calculation or source support

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### Criterion: Limit

**0 points:** Overclaims certainty

**1 point:** Vague limitation

**2 points:** Relevant uncertainty or missing fact

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### Criterion: Transfer

**0 points:** Copies without adapting

**1 point:** Adapts with prompts

**2 points:** Solves a changed case independently

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Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold. Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

## Feedback

Ask the learner to find the evidence that challenges this claim: “Index 1 always selects the first list item.” Keep the first attempt and an uncoached retry separate.

## Original transfer answer

Sum=24, count=3, mean=8.0. For [] the mean is undefined; explicitly report missing/insufficient data before division. Returning zero invents a measurement and can distort later predictions. Test both populated and empty inputs.

## Explanations for the additional workbook examples

### W1 State a schema

List the required fields, types and duration boundaries for the worked table.

activity: nonempty text; minutes: integer from 0 to 60 inclusive.

Check both fields. activity must exist as nonempty text. minutes must be an exact integer and satisfy  $0 \leq \text{minutes} \leq 60$ . Inclusive means equality is accepted at both ends. Structure, type and range are separate checks; passing them does not prove the duration was measured accurately.

### W2 Classify records

Classify minutes values 0, 60, -1 and 61 under this schema.

0 and 60 pass; -1 and 61 fail.

Apply the inclusive range to each integer. Both 0 and 60 are on allowed boundaries. -1 is below the minimum and 61 is above the maximum. Require the reason for each rejection, not only pass/fail labels.

### W3 Repair a unit problem

A source says 180 seconds but the row says minutes: 180. Give a documented correction.

Divide by 60 to record 3 minutes and retain the source value and conversion note.

### W4 Design a schema independently

For a fictional shelf audit, define label and book\_count fields. Require a nonempty text label and an integer count from 0 to 40. Give two valid and two invalid records with reasons.

Examples: A/0 and B/40 valid; empty label/5 invalid label; C/41 invalid range. Text "5" also fails the count type. Accept alternatives meeting the written rules.

### W5 Separate validity and truth

A row says Reading, 12 and passes. Does that establish the session really lasted 12 minutes?

No. The value has a permitted structure but still needs trustworthy collection evidence.

### File input and output — W1 Trace types

Before and after `int("8")`, identify the types and values.

Before: text "8". After: integer 8.

Before conversion, "8" is a string containing the character 8. `int("8")` returns the integer 8, which can participate in numeric arithmetic. The conversion does not change the original file and does not establish that the number meets the duration schema.

## File input and output — W2 Predict the total

A sandbox file has lines 5, 0 and 2. What total does the worked reader print?

The parsed values sum to  $5 + 0 + 2 = 7$ .

## File input and output — W3 Repair a path issue

The file is in the class-data folder but the script looks in its working folder. Give a safe repair.

Use the correct path to class-data/durations.txt or run from the intended folder. Keep read mode and do not overwrite the input.

## File input and output — W4 Read independently

In a fresh sandbox file counters.txt put 4, 6 and 0 on separate lines. Adapt the reader, print the number of values and total. State the expected results.

Read counters.txt as UTF-8 text, convert three lines to integers; count 3 and total 10. Keep the raw file unchanged.

## File input and output — W5 Handle malformed data

The file contains twelve on one line. Why should the program not silently turn it into zero?

It is a parse failure, not a measured zero. Flag the line and report the problem; silently substituting changes the evidence.

## Exceptions and recovery — W1 Trace success

What does `parse_minutes("0")` return, and why is zero allowed?

A record with status ok and value 0; zero meets the inclusive range.

## Exceptions and recovery — W2 Trace two failures

Compare `parse_minutes("abc")` and `parse_minutes("70")`.

abc gives invalid text/None; 70 gives out of range/None. Both fail but for different reasons.

## Exceptions and recovery — W3 Repair a handler

An except block returns 0 for any failure. State two problems and a repair.

It confuses failure with observed zero and may hide unrelated bugs. Catch `ValueError`, return an explicit status and None, then validate range.

## Exceptions and recovery — W4 Build independently

Adapt the function for count strings representing integers 0 through 10. Test "4", "0", "11", "four" and "-1" and record status/value.

4 and 0 give ok with 4 and 0; 11 and -1 give out of range/None; four gives invalid text/None. Use specific handling and explicit range checks.

## Exceptions and recovery — W5 Choose recovery

A batch has 8 valid and 2 invalid lines. What should its report include?

Accepted count 8, rejected count 2 and reasons; summaries apply to accepted data. Preserve raw input for authorized correction.

## Assess independent transfer

Selected case: Supplied values are 3, 7, 14. Compute their mean. What should a function report for an empty list? How should it handle a missing fourth record?

## Selected case answer

Mean =  $(3+7+14)/3=8$ . An empty list has no observed mean; return an explicit unavailable result rather than divide by zero or invent 0. State the missing-value policy and report coverage  $3/4$  separately if one of four records is missing.

### Dimension · Evidence required

**Dimension:** Explanation

**Evidence required:** Names the applicable idea or procedure and ties it to the supplied facts.

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**Dimension:** Trace and result

**Evidence required:** Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.

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**Dimension:** Limits

**Evidence required:** States the assumptions, missing evidence or conditions under which the result would change.

---

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace. Do not penalise an honestly recorded model failure if the learner correctly explains its evidence and limits.

## Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case. Confirm it was not

previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

## 6. Connect and extend learning

### Connect and extend learning

#### Responsible Plant Classifier

Create a tiny five-row plant table with explicit column names, types, units, and a reproducible summary calculation.

Keep the decision, input, result, explanation and remaining limitation in the project evidence record. Retrieve this record next week rather than restarting the project.

#### Support and stretch

Use a tiny labelled dataset, explicit units and a named positive class. Stretch: require a baseline, held-out evaluation and subgroup or shifted-condition analysis with denominators.

#### Exit reflection

Ask for one decision the learner can now justify and one situation requiring more evidence or human review. Confidence ratings are separate from correct independent performance.

#### Bridge to the next week

Next: Week 5: Inspect Before Modeling. Retrieve today's evidence and identify what carries forward and what is new.

#### Support and fresh reassessment

Use a schema card beside each record. Extension: explain how allowing decimal durations would change both type and rounding rules.

Independent W4 evidence: Examples: A/0 and B/40 valid; empty label/5 invalid label; C/41 invalid range. Text "5" also fails the count type. Accept alternatives meeting the written rules. Record whether the learner needed no help, a hint or explanation. Fresh retry: define a tray label and integer count 0 through 20. Require tests at 0, 20, -1 and 21 and a text count.

Next week: Missing values and validation. Reuse this week's explicit contracts and evidence.

#### Readiness and reassessment for File input and output

Provide the exact file and folder path. Extension: retain line numbers alongside parsed values to make corrections traceable.

Independent W4 evidence: Read counters.txt as UTF-8 text, convert three lines to integers; count 3 and

total 10. Keep the raw file unchanged. Record whether the learner needed no help, a hint or explanation. Fresh retry: sandbox values 2, 7, 1, 0 give count 4 and total 10. An empty file yields count 0 and sum 0, which must be labelled no observations.

## Readiness and reassessment for Exceptions and recovery

Use success and failure path cards. Extension: reject non-string input explicitly with a separate status before calling int.

Independent W4 evidence: 4 and 0 give ok with 4 and 0; 11 and -1 give out of range/None; four gives invalid text/None. Use specific handling and explicit range checks. Record whether the learner needed no help, a hint or explanation. Fresh retry: integer score strings 0 through 5. "5" ok/5, "6" out of range/None, "" invalid text/None, "0" ok/0.

## Plan the delayed return

Suggested return points in this level: ai-4/5, ai-4/7, ai-4/10. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used. For a new level, revisit the previous capstone before checking the new prerequisites.

### Record · Teacher evidence

**Record:** Learner code and date

**Teacher evidence:** \_\_\_\_\_

\_\_\_\_\_

**Record:** Concept and case used; prior exposure

**Teacher evidence:** \_\_\_\_\_

\_\_\_\_\_

**Record:** Explanation / trace / limits

**Teacher evidence:** \_\_\_\_\_

\_\_\_\_\_

**Record:** Support and first repair

**Teacher evidence:** \_\_\_\_\_

\_\_\_\_\_

**Record:** Changed case and delayed outcome

**Teacher evidence:** \_\_\_\_\_

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**Record:** Next teaching action

**Teacher evidence:** \_\_\_\_\_

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Use the evidence to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

### Project evidence from this case

For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

## 7. Prepare, teach and assess

Teaching target: Specify a table schema

Prepare the labelled worked-case picture, the matching workbook, fictional input cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

### Session · Plan · Resources

**Session:** Session 1 · Understand · 45 min

**Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

**Resources:** Sections 1–6; record the brief recall in the workbook.

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**Session:** Session 2 · Practise · 45 min

**Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

**Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

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**Session:** Session 3 · Investigate and transfer · 45 min

**Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

**Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.



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Before class, solve the supported practice: Classify minutes values 0, 60, -1 and 61 under this schema.

Reasoned practice answer: 0 and 60 pass; -1 and 61 fail.

Check every learner before discussion: Use the worked case for Rows columns and schemas. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: A row can satisfy every schema rule and still be a wrong observation. Schema validation checks our stated structure and constraints, not whether someone measured honestly or selected a representative sample.

Support: read the exact case aloud, enlarge the labelled picture, supply input cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: S changes from 0 to None. Require a prediction and a reason before comparing. Expected result: Mean becomes  $(2+6)/2=4$  cm with  $2/4$  measured. Reason: Removing a measured zero changes both the denominator and the coverage.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

### Criterion · Not yet · With support · Independent

**Criterion:** Explanation

**Not yet:** Names an answer without a deciding fact

**With support:** Uses a hint to link fact and decision

**Independent:** Explains the deciding fact in own words

---

**Criterion:** Trace or test

**Not yet:** Cannot connect input and result

**With support:** Completes steps with prompts

**Independent:** Traces or tests a changed case accurately

---

**Criterion:** Limits and revision

**Not yet:** Claims more than evidence supports

**With support:** Names a limit after a prompt

**Independent:** States a defensible limit and useful next test

---

Project checkpoint: Frame the need. Name the user, the problem, allowed inputs and a safe simple starting method.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and evidence.

## 8. Use feedback, labs and repair paths

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding evidence rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual outputs; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as evidence. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.